

THE HANDBOOK FOR THE PHD PROGRAM IN COMPUTER SCIENCE

2025-2027 GRADUATE CATALOG

DEPARTMENT OF COMPUTER SCIENCES

THE UNIVERSITY OF TEXAS AT SAN ANTONIO

Revised **Fall 2025**

Table of Contents

<i>Introduction.....</i>	<i>3</i>
<i>Admission Requirements.....</i>	<i>3</i>
<i>Program Guidelines</i>	<i>3</i>
Program of Study	3
Transfer of Credit	4
Advisors	4
The Milestones and Timeline.....	5
Advancement to Candidacy	5
Qualifying Examination	5
Doctoral Dissertation Proposal Examination.....	7
The Dissertation Committee.....	7
Doctoral Dissertation and Final Oral Examination.....	8
Change of Program.....	8
<i>University-Wide Policies.....</i>	<i>8</i>
Enrollment requirements	8
Academic Standing.....	9
Academic Dismissal	9
<i>a. Frequently Asked Questions on Dissertation Proposal Defense.....</i>	<i>10</i>
<i>B. Forms.....</i>	<i>14</i>

INTRODUCTION

The Department of Computer Science offers advanced coursework and research leading to the Doctor of Philosophy (Ph.D.) degree in Computer Science. Successful Ph.D. candidates must demonstrate an in-depth knowledge of computer science and must deliver an original contribution to the field.

ADMISSION REQUIREMENTS

Contact the UTSA Graduate School (<http://graduateschool.utsa.edu>) for application forms by mail or to apply on-line.

The minimum requirements for admission to the Doctoral degree program in computer science in addition to University-wide graduate admission requirements are as follows:

- a B.A., B.S., or M.S. degree in computer science or a related area;
- three letters of recommendation attesting to the applicant readiness for doctoral study.
- the GRE general test verbal, math, and analytical sections. When GRE scores are used to determine an admission, applicants will be compared to applicants with similar socioeconomic backgrounds; **GRE will be waived for applications on/after Spring 2024**

Admission is competitive. Satisfying the minimum requirements does not guarantee admission. An application should also include a resume and a statement of research experience and interest. Applicants will automatically be considered for scholarships, and teaching and research assistantships. To receive full consideration all application materials should be received by **February 1** for Fall admission or **October 1** for Spring admission. A complete application includes the application form, transcripts, three letters of recommendation, a resume, a statement of research experience and interest, the TOEFL (or other accepted tests) score for those applicants whose native language is not English and they have not graduated from a United States institution.

PROGRAM GUIDELINES

These guidelines for the Doctor of Philosophy Degree in the Department of Computer Science at UTSA describe requirements and procedures that supplement, but do not supersede, the Doctoral Degree Regulations listed in the UTSA Graduate Catalog. Candidates for the degree are required to successfully complete a **minimum of 72 semester credit hours** of graduate coursework as described in the program of study.

PROGRAM OF STUDY

- A. Core course (3 semester credit hours):
 - CS 5633 Analysis of Algorithms
- B. Required core electives (9 semester credit hours):
 - CS 5103 Software Engineering
 - CS 5163 Data Sciences

- CS 5233 Artificial Intelligence
- CS 5323 Principle of Cybersecurity
- CS 5363 Programming Languages and Compilers
- CS 5513 Computer Architecture
- CS 5523 Operating Systems

C. Other Electives (12 semester credit hours):

Students must complete at least 12 semester credit hours of additional eligible organized graduate courses in the Department of Computer Science.

D. Other elective and research courses (48 semester credit hours minimum), including:

- CS 7123 Research Method (3 semester credit hours minimum)
- CS 7211-6 Doctoral Research (18 semester credit hours minimum)
- CS 7311-6 Doctoral Dissertation (9 semester credit hours minimum)

With prior approval of the Graduate Advisor of Record, students may apply a **maximum of 6 hours** of graduate courses from other disciplines to the degree.

TRANSFER OF CREDIT

Students may transfer prior graduate study **up to 30 semester credit hours** from another institution toward the Doctor of Philosophy degree in Computer Science with the approval of the Graduate Studies Committee. Each student transcript will be evaluated by the Graduate Studies Committee, and credit will be determined on a course-by-course basis to satisfy the requirements of the degree.

ADVISORS

Upon entering the Ph.D. program, the Computer Science PhD Graduate Advisor of Record (GAR) will be responsible to provide general advising to students based on the Milestones Agreement Form of UTSA Doctoral Program in Computer Science. More important, **each student is expected to find a Tenured/Tenure-Track faculty as the student Doctoral Advisor to start working on research in the first semester.** The Doctoral Advisor will supervise the student on research and work with the student to pass the **PhD Oral Qualifying Examination** no later than the end of the third long semester. After that, the student will be advised by the student's Doctoral Advisor who directs the research of the student, with the goal of passing a dissertation proposal examination and successful completion of a dissertation.

The Doctoral Advisor should be consulted each semester before the student registers for courses. It is also possible for a student to have two Doctoral Advisors, who jointly supervise the student. A Doctoral Advisor must be a member of the Computer Science Graduate Council. A student may change Doctoral Advisor by submitting a Change of Advisor form for approval by the Graduate Studies Committee.

THE MILESTONES AND TIMELINE

The progress of the students in the Ph.D. program is specified on the Milestones Agreement Form of UTSA Doctoral Program in Computer Science. See the form for the expected timeline for students to reach each milestone. Students admitted to the program are required to sign the form and submit it to the Department of Computer Science within the first year of the enrollment into the program.

An approximate timeline for completing the program by a full-time PhD student with a BS degree is as follows.

- Year One: complete CS 5633 Analysis of Algorithms and CS 7123 Research Method; Select and work with a Doctoral Advisor on research projects with Independent Studies; prepare for the PhD Oral Qualifying Examinations.
- Year Two: **Pass the PhD Oral Qualifying Examinations before the end of the 3rd long semester**; register CS 7211-6 Doctoral Research with the Doctoral Advisor every semester. Complete three required elective courses by the end of the year.
- Year Three: Complete all required organized courses; Work on the dissertation proposal. Form a dissertation committee.
- Year Four and Beyond: **Pass the Dissertation Proposal Examination by the end of the 4th year**; Apply for candidacy; Write a dissertation. Pass the Final Oral Examination.

Each student enrolled in the PhD program will be evaluated annually for their progress in the program in terms of those milestones in the Milestone Agreement Form. The Annual Report Form is required to complete the annual evaluation and to file the progress report to the Graduate School.

Students whose progress in the program are less than satisfactory as compared to the Milestone Agreement will be required to provide an action plan to address the problems that cause the delay of the progress.

ADVANCEMENT TO CANDIDACY

Students seeking a doctoral degree must be admitted to candidacy. The requirements for admission to candidacy include passing a Doctoral Qualifying Examination and a doctoral dissertation proposal examination. Students should consult the University's Doctoral Degree Regulations for other requirements.

QUALIFYING EXAMINATION

To pass the PhD Qualifying Examination in Computer Science, a computer science doctoral student must meet the following requirements.

1. **QE Oral presentation requirement:** pass the Oral Qualifying Examination, which focuses on written and oral research capability requirements, by the end of the 3rd long semester;
2. **GPA requirement:** A GPA of 3.3 or better in CS 5633 Analysis of Algorithms and any three (3) of the required elective courses

The Oral QE committee should have three members, including the doctoral advisor of the student. All the members should be PhD-advising faculty members (Tenure/Tenure Track) from the CS department. The non-advisor members should be proposed by the student with the approval from the advisor and approved by the PhD GAR who may replace one of the proposed members with another eligible faculty member. If the PhD GAR is the advisor, the non-advisor members should be approved by the department chair who may replace one of the proposed members with another eligible faculty member.

A student may present either a literature review in a research topic chosen by the advisor, ongoing research work, or an accepted / published paper. The assessment should evaluate (1) whether the student can synthesize a collection of existing works into an assessment of the state-of-the-art on a problem; and (2) whether the student can critically analyze research work, including identifying weaknesses, open problems, and opportunities for further work.

For a student to pass the oral presentation part of the qualifying exam, all three committee members must agree unanimously. Students who fail the qualifying exam should be instructed about what needs to be done to pass the exam and have one additional chance to retry within 60 days with the same or a different committee.

If a student plans to do an oral presentation in a certain semester, the student needs to report to the PhD GAR by the 2nd week of that semester. The oral presentation committee needs to be proposed by the end of the 5th week of the semester, and the oral presentation needs to be held before the 10th week of the semester.

Students transferred from other universities with their advisors may petition to the graduate studies committee to waive the oral presentation based on the qualifying exam they passed in their previous universities.

Full-time doctoral students must pass the PhD Oral Qualifying Examination by the end of the third long semester in the doctoral program.

Part-time doctoral students must (a) meet the GPA requirement by the time they complete the first 18 hours of computer science courses at UTSA while enrolled in the doctoral program, and (b) meet all requirements of Qualification Examination prior to attempting more than 18 accumulated hours of computer science courses while enrolled in the doctoral program. Failure to meet these requirements by the given timeline will normally result in dismissal from the program.

GPA Verification

For students who take the core and required elective courses (CS 5633 Analysis of Algorithms and other three courses) at UTSA, the most recent course grades obtained will be used to verify the GPA requirements. For students who transfer the courses for credit, the grades (or grades converted to equivalent UTSA grades) indicated in their admission transcripts will be used to verify the GPA requirements. **A student should not transfer CS 5633 Analysis of Algorithms.**

DOCTORAL DISSERTATION PROPOSAL EXAMINATION

After a student has passed the qualifying examination and has made progress in doctoral research, the next step is the Doctoral Dissertation Proposal. The student has to prepare a written proposal for a dissertation topic, and form a Dissertation Proposal Committee chaired by the student's doctoral advisor with **at least three additional graduate faculty members** in Computer Science Department. The Dissertation Proposal Committee will conduct an oral examination during which the student presents the dissertation proposal. The presentation is followed by a period of questioning based on the dissertation proposal. Unanimous approval of the Dissertation Committee is required to pass the oral examination. No more than two attempts to pass the oral examination will be permitted.

The student should submit the dissertation proposal to the students Dissertation Committee at least two weeks prior to the examination. A Program of Study form must also be submitted by the student and be approved by the Graduate Studies Committee at this time. The dissertation proposal should:

- present an overview of the background and related work in the field,
- explain the basic idea of the dissertation topic,
- argue why that topic is original, challenging, and important,
- state what kind of results are expected, and present preliminary results, if any, and
- make a plausible argument that these results are obtainable within a reasonable amount of time.

The student should write the dissertation proposal as soon as they can address the issues described above.

The answers to a set of Frequently Asked Questions to the dissertation proposal examination is in **Appendix A**. Note that if after passing the dissertation proposal examination, the student changes a Doctoral Advisor and starts a different research topic under the supervision of the new Doctoral Advisor, the student must pass another dissertation proposal examination on the new research topic.

After a student has passed the doctoral dissertation proposal examination, the student must register for CS 7311-6 Doctoral Dissertation every semester until the student completes the degree.

THE DISSERTATION COMMITTEE

The Dissertation Committee of a student consists of a minimum five faculty members, with the students Doctoral Advisor being the chair of the committee. The remaining members of the committee should be selected by the student, in consultation with the students Doctoral Advisor. The chair and three of the other members of the committee must be members of the Department of Computer Science. One remaining member should be from outside of the Department of Computer Science. All the members of the Dissertation Committee must be Members or Special Members of the UTSA Graduate Council. A committee member from outside the University can become a Special Member of the Graduate Council with approval by the Computer Science and the UTSA

Graduate Council. The composition of the committee must be approval by the Graduate Studies Committee, the College of Sciences, the Graduate School, and the Provost and the Vice President for Academic Affairs.

DOCTORAL DISSERTATION AND FINAL ORAL EXAMINATION

After a student has passed the doctoral dissertation proposal examination, the next steps are writing a dissertation and passing the final oral examination. The final oral examination is administered and evaluated by the students Dissertation Committee and covers the dissertation and the general field of the dissertation. The final oral examination consists of an open presentation of the dissertation followed by an oral examination. Unanimous approval of the Dissertation Committee is required to pass the final oral examination. Also, the dissertation must be unanimously approved by the Dissertation Committee.

It should be emphasized that the research presented in the Dissertation must be included in **at least one paper** that is **published or accepted** for publication in a peer-reviewed journal or conference/workshop before a Ph.D. student graduates.

CHANGE OF PROGRAM

A student who wants to exit the Ph. D. program may apply to switch to the Masters Degree program in computer science. The student may apply for a M.S. degree in Computer Science by satisfying all requirements of the M.S. degree. All the graduate courses taken as a doctoral student **except Research Seminar and Doctoral Research** may be counted towards a M.S. degree.

A student who wants to switch to the Master Degree Programs must petition for a change in status from Ph.D. degree seeking to M.S. degree seeking. The petition must be approved by the Graduate Studies Committee prior to the deadline for applying for graduation.

A student who is supported by a stipend, a teaching or research assistantship automatically rescinds the financial support and any tuition waiver upon changing status to an M.S. seeking student. A student who applies for the interim M.S. degree after being admitted to Ph.D. candidacy retains the status as a Ph.D. seeking student.

UNIVERSITY-WIDE POLICIES

ENROLLMENT REQUIREMENTS

A full-time student must take at least 9 hours per fall or spring semester. Full-time enrollment during the summer semester is 3 credit hours. International graduate students must be full time to maintain F-1 visa status (see

<http://international.utsa.edu/current-students/enrollment/>). Summer enrollment for international students is optional as long as they are eligible and intend to register for the following fall semester. However, summer enrollment is required if the student is a teaching or research assistant.

ACADEMIC STANDING

Good academic standing requires a minimum GPA of 3.0. Student are placed on academic probation if:

- 1) The GPA is below 3.0 in any semester at UTSA. The GPA is based on all courses taken so includes graduate and undergraduate levels.
- 2) A grade of D+, D, or D- is earned in any course in a semester.
- 3) On reinstatement to the university following an academic dismissal.

ACADEMIC DISMISSAL

A student who meets any of the following conditions is placed on the academic dismissal list:

- 1) Admission conditions are not met.
- 2) A grade of F is earned in any course.
- 3) The GPA is below 2.0 in any semester.
- 4) Conditions leading to academic probation occur in the semester following placement on academic probation.
- 5) Failure to pass an oral or written exam (such as Comprehensive Examination) required for the degree after the maximum of two attempts.

If dismissed from the university, a petition application for reinstatement can be filed after one long semester.

A. FREQUENTLY ASKED QUESTIONS ON DISSERTATION PROPOSAL DEFENSE

A.1 What is a proposal defense?

A proposal defense, which consists of an oral presentation and an oral examination, is one of the steps designed to prepare you to become an independent researcher. During the proposal defense you will explain the general problem or research area of your work. You will demonstrate your understanding of the background material in this area and your general grasp of how the problem you have picked fits into the big picture. Before your proposal defense, you will submit a written dissertation proposal to your proposal committee. (See Question 8 for details.)

A.2 What is the format of the proposal defense?

The proposal defense consists of a presentation (50 minutes), a question period, and a closed session with the committee. The audience may stop and ask you questions at any time, so often the presentation runs longer. You should time your presentation to be 40 to 50 minutes if there are no questions. Here is advice for the format of the presentation:

1. Since you only have 40-50 minutes, you should cover elementary background material quickly.
2. It is essential that you get to substantive topics in your allotted time.
3. You must be ready to answer elementary questions on the background material of your area. You should also be prepared to discuss related work and to describe how your work is different.
4. You must prepare your talk carefully, making sure you have the formulas and analysis correct. Only the content of this talk really matters, and not any particular flashy format.
5. It is a good idea to put small examples to the side to show that you understand any formulas and can explain them.
6. You want to convince the committee that you clearly understand what you are doing.
7. Emphasize your research contributions and future directions.
8. The material of your presentation should reflect your written dissertation proposal, which is described in more detail below.
9. You should practice your presentation prior to the defense, ideally with some friends who can ask you questions about things that are not clear.
10. Your advisor should always review the proposal before you submit it to your committee members.
11. Please proofread your dissertation proposal and slides carefully.

A.3. What kinds of questions will my committee ask me?

One of the purposes of the proposal defense is to demonstrate that you understand the problem that you are working on. Remember that sometimes your committee members will ask you questions, not because they want to know the answer, but because they want to know if you do. They will question the meaning of formulas, your graphs, the assumptions, and especially your contribution and your ideas.

A.4. What is the composition of the committee?

The committee consists of your advisor and at least three faculty members from computer science. Usually this committee ends up as part of your dissertation committee, but that is not required. You should pick your committee in conjunction with your advisor to represent expertise that might be useful to you in completing your dissertation. Your committee must be approved by the PhD doctoral studies committee.

A.5 When should I do my proposal defense?

Generally you should try to do your proposal defense in your third year.

A.6 How is the proposal defense related to other exams?

There are three major exams that you must pass to earn a PhD in CS. These exams measure your progress towards becoming an independent researcher.

The first exam is the written exam taken at the end of the first year. This exam verifies your competency in core subjects. When you pass this exam, you can sign up for doctoral research.

The second exam is the proposal defense, which is also the CS PhD oral examination. The purpose of the proposal defense is to make sure that you understand the problem and have the necessary background to do research in the area you have selected. The exam also establishes that you have a game-plan for your research. When you pass this exam, you advance to candidacy and can sign up for doctoral dissertation.

The final exam is the actual dissertation defense. At this exam you explain and defend your original research as embodied by your thesis document and publications. You set your work and contributions in the bigger picture of the field.

A.7 What should I do to prepare for the proposal defense?

1. Pick a proposal committee and get the committee approved
2. Prepare your dissertation proposal and submit it to your committee at least 2 weeks before the defense. (Ideally you should do this 4 weeks before the defense and then meet individually with your committee members 2 weeks before to discuss any changes that need to be made in the dissertation proposal before the defense. You should also give your committee copies of the papers that you have co-authored.)
3. Prepare your presentation
4. Schedule the defense with the Department office and your committee members. You should allow a 2 hour block of time for this.
5. Practice your presentation
6. Make any corrections or complete other work required by your committee

A.8 What should be in the dissertation proposal?

A written dissertation proposal should be submitted to the student's Oral Examination Committee at least two weeks before the presentation. The guidelines state that the dissertation proposal should:

1. Present an overview of the background and related work in the field.

2. Explain the basic idea of the dissertation topic.
3. Argue why that topic is original, challenging, and important.
4. State what kind of results are expected and present preliminary results, if any.
5. Make a plausible argument that these results are obtainable within a reasonable amount of time.

The proposal should also contain a clear description of your contribution to the publications and work done jointly so far. You should also clearly delineate in both your presentation slides and in the dissertation proposal which results are yours and which results you are using from other work (with a reference).

A. 9 How many references are enough?

The reference list for the proposal may not include all of the references that end up in your dissertation, but it is important that you have found the relevant related work so that you don't duplicate results for your dissertation. You include recent references and references that are from high-quality venues. Part of learning to become an independent researcher is to learn what the high-quality publications are and to recognize high-quality work. If you have any questions, ask your advisor or committee members. You should not have references that are mostly 5 years old or are all less selective conferences. At least some of your references should be from the top journals or conferences in that area.

A.10 Will my proposal defense really take 2 hours?

Proposal defenses usually run 1 to 2 hours. Plan on 2 hours.

A.11 What if I change my topic after I pass the proposal defense?

You do not have to pass the defense again if you make minor changes in your topic or the plan to complete your dissertation, but you need to keep your committee and the Graduate Advisor of Record informed. The Graduate Advisor of Record in conjunction with the Graduate Studies Committee will decide whether the change is major enough to warrant redoing the dissertation proposal or changing the membership of your dissertation committee.

A.12 What is the role of the advisor versus the committee?

Your advisor plays the most important role in guiding your research, and will generally be the one to tell you when to do your proposal defense and your dissertation defense. However, your committee as a whole has to pass you on these milestones. They can be a valuable resource in suggesting directions. They also are likely candidates for recommendation letters when you graduate. You should keep them informed of your progress.

A.13 What does the PhD degree mean?

Earning a PhD signifies that you are an independent researcher who has made original contributions to your field. Even seemingly modest results can take on more importance later.

A.14 How soon/long is it between the proposal defense and the dissertation defense?

A student needs to register for Doctoral Dissertations for at least two semesters. However, the key to earning a PhD is to make an original contribution. As such there can be no fixed time limit. A typical time is one to two years, but this varies considerably depending on student and research area.

A.15 How do I know when I am becoming an independent researcher?

Becoming an independent researcher doesn't happen overnight. You will begin by working with your advisor and possibly a small group of other students on research projects, often in a supporting role.

As you mature, you should be contributing more of the ideas and doing more of the writing. Research quality is generally judged by publication quality. As your contribution to the research increases, generally your position in the authorship moves forward. (Some theoretical fields always publish alphabetically, but most go by contribution, with the advisor going either first or last in the list.) Your first paper will probably not get into a great venue unless someone else does the writing. As you get more experience and the quality of your work increases, you should be targeting better conferences and journals. It is important to make a significant enough contribution to the work that constitutes your thesis that you are first author on the publications reporting that work. However, you can't expect to be first author if you do not make this significant contribution.

B. FORMS

This appendix contains the following forms.

1. Milestone Agreement Form
2. PhD Annual Report Form
3. Selection of Doctoral Advisor Form
4. Change of Doctoral Advisor Form
5. Appointment of Oral Examination Committee
6. Completion of Qualifying Exam Form
7. Dissertation Proposal Approval Form
8. Appointment of Dissertation Committee Form
9. Application for Candidacy Form
10. Certification of Completion of Dissertation Form
11. Program of Study Form
12. Petition for Transfer Credits form